

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fourth Semester of B.Tech. (IT) Examination****May 2022****IT254 Computer Architecture & Microprocessor Interfaces****Date: 09.05.2022, Monday.****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION – I**Q - 1 Answer the Following Questions.****[06]**a. Which of the Following Generation of Computer is Based on Integrated Circuit? **[1]**

i) First Generation

ii) Second Generation

iii) Third Generation

iv) Fourth Generation

b. Which of the Following is Representation in Signed 2's Complement of $(-35)_{10}$? **[1]**

i) 1011101

ii) 1100011

iii) 1010011

iv) 1101110

c. The Exponent Bias for Single Precision IEEE 754 is _____. **[1]**

i) 127

ii) 128

iii) 1023

iv) 1024

d. State True or False: Instruction size in CISC architecture is fixed. **[1]**e. "Pipeline Improves Throughput of the System." Justify the Statement. **[2]****Q - 2 Answer the Following Questions. (Any Three)****[15]**a. Explain Any Five Addressing Mode of MIPS ISA. **[5]**b. (i) If clock rate is 50 MHz, find execution time for a program with 1000 instructions, if CPI for program is 3.5. **[5]**

(ii) What if clock rate increases from 50MHz to 250 MHz and other functions remain same, how many times faster will computer be?

c. Convert the Following Code into Machine Code. (Assume Suitable Data) **[5]**

IF (A == B)

I = J + K;

ELSE IF (A < B)

I = J - K;

ELSE

I = J AND K;

d. Consider a Given Scenario: **[5]**

	A (1 Cycle)	B (3 Cycles)	C (5 Cycles)
Compiler 1	10	7	2
Compiler 2	4	2	6

Assume the Instructions Are in Billions and Frequency is 200 MHz. Find Out CPU Time of Compiler 1 & Compiler 2.

- Q - 3 Answer the Following Questions. (Any Two)** [14]
- List Major Pipeline Hazards. Elaborate Any Two in Detail. [07]
 - Explain Arithmetic and Logical Instructions Set. [07]
 - Differentiate RISC & SISC Architecture. [07]

SECTION - II

- Q - 4 Answer the Following Questions.** [06]
- In the Processor, _____ Transfer is not Possible. [1]
 - Register to Register
 - Register to Memory
 - Memory to Memory
 - None of the Mentioned
 - In the Register File, _____ is Used to Store the Data in the Desired Address. [1]
 - WAD (Write Address)
 - WD (Write Data)
 - None of the Mentioned.
 - All of the Mentioned
 - What will be the Value of $X + Y$, if $X = 00111001$ & $Y = 01001000$? (Here X & Y are Signed Numbers) [1]
 - 00000001
 - 10000001
 - 10000101
 - Overflow
 - In Booth's Multiplication, _____ Shift is Required. [1]
 - Logical Left.
 - Arithmetic Left.
 - Logical Right.
 - Arithmetic Right.
 - To Transfer 25K Data, _____ Bits (Address Lines) are Required. [1]
 - 14
 - 15
 - 24
 - 25
 - In the _____ Circuit, The Output Depends Not Only Present Input but also The Past Input. [1]
 - Combinational
 - Sequential
 - None of the Mentioned.
 - All of the Mentioned

- Q - 5 Answer the Following Questions. (Any Three)** [15]
- Explain IEEE 754 With Suitable Example. [5]
 - Explain the Steps of Instruction Execution. [5]
 - Explain Memory Hierarchy Organization. [5]
 - Explain Arithmetic Pipeline with Example. [5]

- Q - 6 Answer the Following Questions. (Any Two)** [14]
- Draw Data Path of CBZ (LEGV8) / BEQZ (MIPS) Instruction. [07]
 - Explain 2-Way Set Associative Cache Memory with Diagram. [07]
 - Compute $(-9) \times (-13)$ using Booth's Algorithm. Show the Intermediate Steps of it. Consider 5 Bit Representation. [07]

*** Best of Luck ***